

Combined SoC Base Specification

Combined SoC Features, Combined Data Path and Combined Control Path

This document merges three SoC projects into one reference for the combined SoC testbench and the overall SoC structure. Every feature, data path and control path from the three source documents is carried over. Where sources overlap, entries are merged and the sources are listed. Where they conflict, both versions are kept and the conflict is listed in Appendix A.

Tag	Source project	Main contribution
CE	CARDIOEDGE	VeeR EL2 core, AXI4 interconnect, SPI ADC interface with FIFOs, 31-tap FIR with notch banks, QRS accelerator with arrhythmia flag, SPI2, rich GPIO and Timer.
SS	SentinelSoC	RV32IM core, AXI4-Lite memory map, on-chip SAR ADC, 16-tap FIR, hardware alert, AES-128, Watchdog, Interrupt Aggregator, error handling.
SA	RISC-V Spectrum Analyzer SoC	FFT accelerator, FFT Memory Interface, FFT Control, host spectrum workflow over UART.

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Appendix A. Integration notes and open decisions; Appendix B. Verification targets carried over

Reading note: a feature is described as implemented only when merged RTL and test results show it. This document lists intended capabilities; no implementation status is claimed. Original IDs (for example CE DP01, CE CP12) are kept in the Src column for traceability.

1. Combined SoC Features

1.1 Combined IP and block inventory

Addresses 0x4000_0000 to 0x4001_6000 for IPs 5-8, 11, 12, 14-16 come from SentinelSoC. Addresses marked proposed use free slots in that map and are not defined by any source.

#	IP / Block	Base address	Role in combined SoC	Src
1	RISC-V CPU	n/a (bus master)	Single bus master that runs firmware. Sources name it VeeR EL2 (CE), RV32IM core (SS) and RISC-V CPU (SA). Core selection is an open decision (see Appendix A).	CE, SS, SA
2	IMEM (instruction memory)	Local to core	32 KB instruction memory (SS). Stores firmware instructions (CE).	CE, SS
3	DMEM (data memory)	Local to core	32 KB data memory (SS). Stores stack, variables and runtime buffers (CE).	CE, SS
4	AXI Interconnect	n/a	Address decode, arbitration, channel and response routing (CE). Each IP sits in its own 4 KB slot; unmapped or unaligned access returns SLVERR (SS).	CE, SS, SA
5	UART	0x4000_0000	Serial link to host: telemetry, debug, commands, results.	CE, SS, SA
6	GPIO	0x4000_2000	Digital input/output, pin-change interrupts, LED/buzzer control and other pin functions.	CE, SS, SA
7	Timer	0x4000_4000	Sample-rate tick, periodic/one-shot timing, compare and capture.	CE, SS, SA
8	Watchdog	0x4000_6000	Pre-timeout IRQ, then chip reset on firmware hang.	SS
9	SPI2 (generic 8-bit SPI master)	0x4000_8000 (proposed, free slot)	Application-result reporting over SPI.	CE
10	FFT Control	0x4000_A000 (proposed, free slot)	Start command and busy/done status registers for the FFT accelerator.	SA
11	FIR Filter (with coefficient banks)	0x4000_C000	Fixed-point FIR with programmable coefficients, BASE / 50 Hz notch / 60 Hz notch banks and bypass.	CE, SS, SA
12	ADC Interface Controller	0x4000_E000	Acquisition front end: SPI-ADC capture (CE), 12-bit SAR on analog_in (SS), digital-sample interface to an external ADC (SA).	CE, SS, SA
13	FFT Memory Interface	0x4001_0000 (proposed, free slot) or separate bus segment	Memory-access path for FFT input/intermediate/output data. Exact protocol and segment depend on the final RTL.	SA
14	QRS Heartbeat Accelerator	0x4001_2000	QRS detection, RR interval, heart rate (BPM), arrhythmia flag, hardware alert.	CE, SS
15	AES-128 Engine	0x4001_4000	ECB and CTR encryption of telemetry blocks.	SS
16	Interrupt Aggregator	0x4001_6000	Collects interrupt sources into CAUSE / MASK and presents one line to the core.	SS
17	FFT Accelerator	Data via FFT Memory Interface	Hardware Fast Fourier Transform on a block of samples.	SA
18	ADC FIFO (32 x 12)	Inside ADC Interface	Buffers ADC samples; blocks overwrite when full and reports overrun.	CE
19	FIR Input FIFO (8 x 16)	Inside FIR path	Buffers firmware-written samples before FIR processing.	CE
20	FIR-to-QRS FIFO (32 x 16)	Dedicated hardware stream	Valid/ready stream from FIR to QRS with no AXI mediation.	CE
21	External ADC / analog source	Off-chip	External converter and analog input (sensor, audio, vibration, ECG/PPG).	SA, SS

1.2 Combined feature / use-case table

ID	Feature / Use case	Required IP / Block	Functionality / Purpose (merged)	Key output / Result	Src
A. Processing, memory, interconnect and supervision					

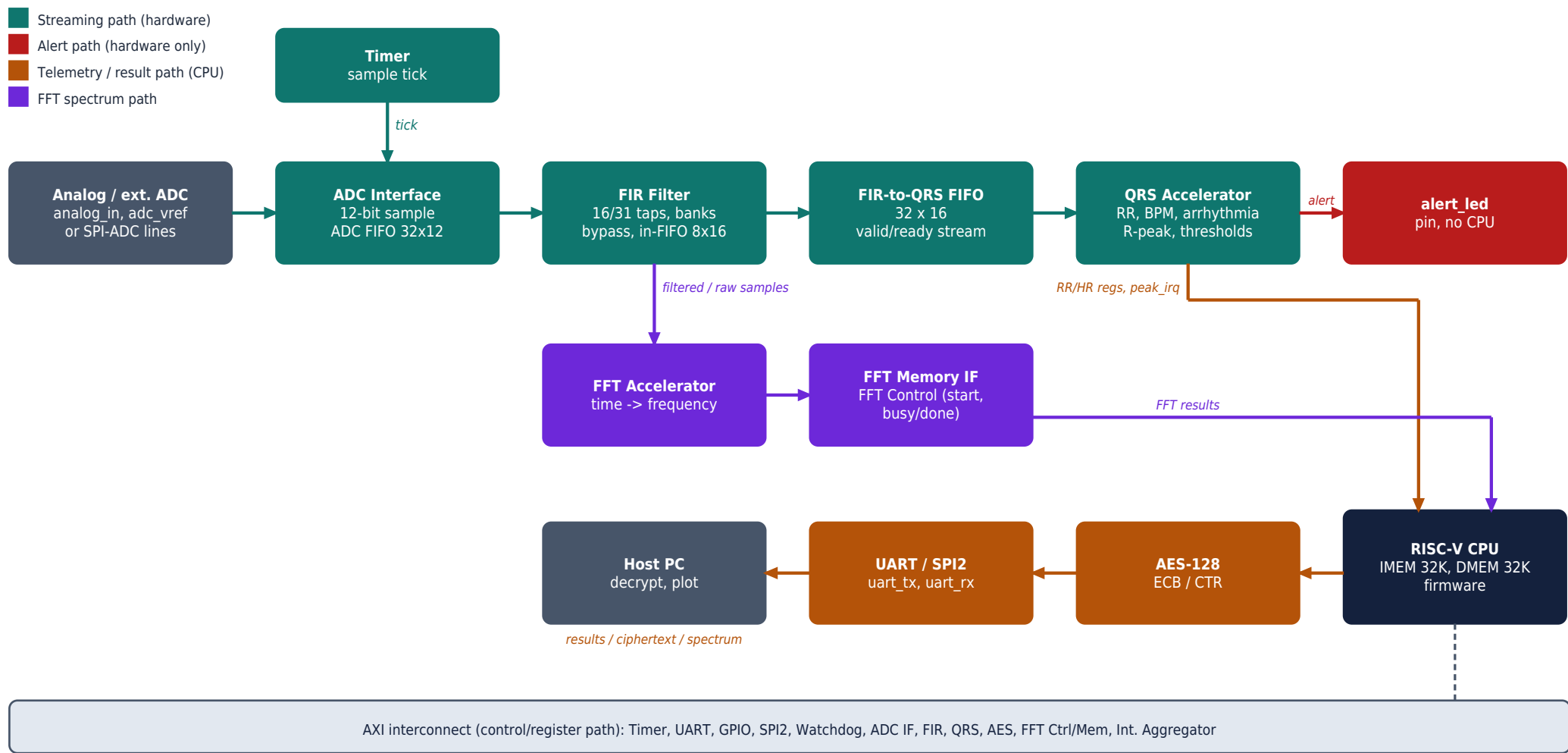
ID	Feature / Use case	Required IP / Block	Functionality / Purpose (merged)	Key output / Result	Src
F01	Run application firmware	RISC-V CPU + IMEM + DMEM	Executes firmware; stores instructions, stack, variables and runtime buffers; controls the SoC. 32 KB IMEM and 32 KB DMEM. Configures and coordinates the peripherals and DSP accelerators (software-based system control).	System control and firmware execution	CE, SS, SA
F02	Connect processor to SoC	AXI Interconnect	AXI address decode, arbitration, channel routing and response routing between the processor and system targets. CPU is the only master in the SS base, so no arbitration is exercised. Each IP is a block of registers in its own 4 KB slot. Anything outside the mapped slots returns SLVERR instead of hanging the system.	Processor-to-memory/peripheral communication	CE, SS, SA
F03	Aggregate interrupts	Interrupt Aggregator	6 sources set bits in CAUSE (all pending sources visible); MASK gates each source (resets to 0x3F); enabled sources merge into one line to the core; write-1-to-clear. Fixed service priority: Watchdog, QRS, Timer, AES, UART, GPIO.	Single interrupt line + CAUSE register	SS
F04	Supervise firmware (watchdog)	Watchdog	Programmable timeout; FEED write restarts the countdown; pre-timeout interrupt as a warning; wdt_reset_n resets the chip on a hang; works even if the CPU is frozen; fed from the timer ISR.	Pre-timeout IRQ, then wdt_reset_n	SS
B. Timing and signal acquisition					
F05	Generate sampling event / timing	Timer	Generates the periodic acquisition event from the 50 MHz system clock (250 Hz ECG sampling). 32-bit counter with load/reload, prescaler and compare; periodic or one-shot mode; compare-match interrupt; live COUNT read-back; capture register stores the timer value on a capture event. Each compare match starts one ADC conversion. Also provides timing or periodic events for software.	250 Hz acquisition timing; periodic/one-shot timing; capture value	CE, SS, SA
F06	Acquire analog / ECG signal	ADC Interface Controller (+ external ADC where used)	CE: controls the ADC SPI interface, captures serial conversion data and assembles the 12-bit sample. SS: 12-bit SAR conversion on analog_in with adc_vref as reference, programmable sample-rate divider, START then EOC flag, OVERRUN flag if the FIR is busy. SA: an external ADC converts the signal; the ADC Interface receives and transfers digital samples into the SoC and may provide sample-valid, status and interrupt. Describe it as a sample interface unless the RTL includes conversion hardware.	12-bit digital sample (sample_data + sample_valid)	CE, SS, SA
F07	Buffer acquired samples	ADC FIFO (32 x 12)	Buffers ADC samples between acquisition and processor servicing; prevents overwrite when full and reports overrun. Also supplies samples to the processing blocks in the required order.	Ordered sample buffering	CE, SA
C. Digital filtering					
F08	Prepare samples for FIR	FIR Input FIFO (8 x 16)	Buffers samples written by firmware before they enter the FIR processing stage (firmware-fed mode).	Samples ready for FIR processing	CE
F09	Filter / condition the signal	FIR Filter	CE: 31 taps, Q1.15 coefficients, 40-bit accumulation, saturated 16-bit output. SS: 16 taps, 12-bit in / 16-bit out, programmable coefficients. SA: programmable FIR attenuates out-of-band noise before FFT (optional, configuration dependent). Removes baseline wander and mains noise.	Filtered signal (16-bit filtered_data + filtered_valid)	CE, SS, SA
F10	Suppress 50/60 Hz interference	FIR Coefficient Banks	Selects the BASE, 50 Hz notch or 60 Hz notch coefficient bank; coefficients are also fully programmable through the register interface.	Selectable signal-conditioning mode	CE, SS
F11	FIR bypass and flow control	FIR Filter	Bypass mode; valid/ready handshake with backpressure so a slow stage pushes back instead of losing data silently; stall flag for a stuck pipeline.	Bypass option; stall flag	SS
F12	Stream filtered ECG to QRS	FIR-to-QRS FIFO (32 x 16)	Dedicated valid/ready hardware stream from FIR to QRS without AXI4 mediation. After one-time CPU setup the ADC-FIR-QRS path runs without CPU involvement.	Continuous filtered-sample transfer	CE, SS
D. Cardiac analytics and alert					
F13	Detect heartbeat / QRS	QRS Heartbeat Accelerator	Pan-Tompkins-style pipeline: derivative, squaring, 32-sample integration, adaptive thresholding and refractory control; finds each R-peak.	Beat / QRS detection event (R-peak)	CE, SS

ID	Feature / Use case	Required IP / Block	Functionality / Purpose (merged)	Key output / Result	Src
F14	Calculate RR interval	QRS Heartbeat Accelerator	Maintains beat history and calculates the interval between valid detected beats. RR/HR registers are double-buffered so the CPU never reads a half-updated value.	RR_INTERVAL	CE, SS
F15	Calculate heart rate	QRS Heartbeat Accelerator	BPM = 15000 / RR_INTERVAL (integer relation, CE). Heart rate is compared with programmable low and high bpm thresholds (SS). QRS rejects a configuration where LOW >= HIGH.	HEART_RATE / BPM	CE, SS
F16	Detect arrhythmia	QRS Heartbeat Accelerator	Compares each new RR interval with the running RR average and asserts the flag when deviation exceeds a threshold (CE). Heart-rate crossing of low/high threshold also raises the alert condition (SS).	ARRHYTHMIA_FLAG; threshold alert flag	CE, SS
F17	Hardware alert (CPU-independent)	QRS Heartbeat Accelerator -> alert_led pin	When heart rate crosses a threshold the QRS block sets its alert flag and alert_led follows directly. No interrupt, no firmware; works even if the CPU is halted. alert_led is not an interrupt and is not a GPIO pin.	alert_led pin	SS
F18	Report each peak to CPU	QRS Heartbeat Accelerator + Interrupt Aggregator	On each new peak, peak_irq reaches the CPU through the aggregator; CPU reads RR and HR over the bus; the interrupt is cleared by reading RR_INTERVAL.	peak_irq + RR/HR registers	SS
F19	Generate firmware alert	GPIO + CPU firmware	Firmware responds to arrhythmia events by controlling LED and buzzer outputs.	LED / buzzer alert	CE
E. Spectrum analysis (FFT subsystem)					
F20	Frequency spectrum analysis	FFT Accelerator	Transforms a block of time-domain samples into frequency-domain complex bins (real and imaginary), computed in dedicated hardware instead of CPU software.	Frequency-domain data	SA
F21	FFT data storage and access	FFT Memory Interface	Provides the memory-access path for FFT input, intermediate and output data. Exact storage arrangement depends on the implemented RTL. Documented separately from the control interface.	FFT data/result access	SA
F22	FFT processing control	FFT Control	Start bit and status registers (busy, done) coordinate FFT execution. A start command is not proof that results are available; CPU must respect busy/done.	Start command; busy/done status	SA
F23	Integrated signal-processing workflow	CPU, interconnect, ADC IF, FIR, FFT Control, FFT Memory IF, UART	Coordinates acquisition, filtering, FFT execution, memory access and communication of results to the host. CPU may compute magnitudes from real/imaginary bins.	End-to-end spectrum result at host	SA
F24	Host-side spectrum display (not SoC hardware)	Host PC software	Host software decodes received bins/magnitudes and plots the spectrum. Listed as a host-side feature, not an SoC feature, unless a display subsystem is built in hardware.	Spectrum plot on PC	SA
F. Security					
F25	Encrypt data (AES-128)	AES-128 Engine	128-bit key and 128-bit block; ECB and CTR modes (with IV); START, BUSY and DONE; aes_done_irq; ignores START, MODE and IV writes while BUSY; register interface only (DATA_IN0-3, DATA_OUT0-3).	Ciphertext block	SS
F26	Secure telemetry	CPU + AES-128 + UART	On each peak the CPU reads RR and HR, packs a 128-bit block, loads key/mode/IV and block into AES, starts it, waits for DONE, reads DATA_OUT0-3 and writes the 16 bytes to UART waiting for TX_READY. Data is encrypted before it leaves the chip.	Encrypted telemetry over UART	SS
G. Communication and I/O peripherals					
F27	Host / debug communication	UART	Byte-oriented full-duplex 8-bit link; programmable baud (16-bit divisor); TX-ready and RX-ready flags; RX-ready interrupt; sticky framing and overrun errors. Carries application/debug status, encrypted telemetry, spectrum results and host commands.	UART status / result / telemetry output; host command input	CE, SS, SA
F28	Report results through SPI	SPI2	Generic 8-bit SPI master for application-result reporting, with control, clock, slave-select and configuration registers and interrupt support.	SPI2 serial result output	CE

ID	Feature / Use case	Required IP / Block	Functionality / Purpose (merged)	Key output / Result	Src
F29	Digital I/O, status indication and pin functions	GPIO	SS: 8 pins, direction set per pin, output register, live input read, per-pin pin-change interrupt, mask and pending flags. CE adds: DATA_I input read, interrupt enable/type/polarity/any-edge, set/clear/toggle/write-mask output control, debounce, inversion, open-drain, pull-up/pull-down and PWM. Used for LED/buzzer, status and control signals.	Pin input/output, GPIO interrupts, PWM	CE, SS, SA
H. Chip-level protection and external interface					
F30	Fault protection and error reporting	Bus, ADC, FIR, QRS, UART, AES	SLVERR for unmapped or unaligned addresses; read-only writes flagged; sticky error flags on ADC, FIR, QRS and UART; QRS rejects LOW >= HIGH; AES interlocks; FIFO overrun reporting.	Error flags / SLVERR	SS, CE
F31	Streaming path and telemetry path split	System architecture	Streaming path (Timer, ADC, FIR, QRS) runs in hardware; telemetry path (CPU, AES, UART) runs over the bus. The hardware alert does not depend on how busy the software is.	Hardware alert + secure telemetry	SS
F32	External pins	Chip boundary	clk, rst_n, uart_tx, uart_rx, gpio[7:0], alert_led, analog_in, adc_vref, sar_clk, wdt_reset_n, ext_irq[n:0]. Also needed from CE/SA: ADC SPI pins and SPI2 pins (names to be defined) and LED/buzzer outputs through GPIO. sar_clk (pin or internal) is still open.	Complete pin list	SS, CE, SA

2. Combined Data Path

2.1 Combined data-flow diagram



2.2 End-to-end flows

Flow 1 - Streaming heartbeat path (hardware, no CPU in the loop)

Timer tick (DP06) -> ADC Interface (DP07) -> FIR (DP09, DP12) -> FIR-to-QRS FIFO (DP13) -> QRS (DP15) -> alert_led (DP17). The CPU only configures it once: sample rate, coefficients and thresholds.

Flow 2 - Firmware-serviced / buffered path

ADC FIFO (DP08) -> CPU -> FIR Input FIFO (DP10) -> FIR (DP12). QRS results are read back by the CPU (DP16) and may drive LED/buzzer through GPIO (DP18).

Flow 3 - Secure telemetry path

peak_irq -> CPU reads RR/HR (DP16) -> AES plaintext (DP22) -> AES ciphertext (DP23) -> UART TX (DP24) -> host PC (DP26). SPI2 (DP27) is the alternate result port.

Flow 4 - Spectrum analysis path

External analog signal -> external ADC -> ADC Interface (DP07) -> ADC FIFO (DP08) -> optional FIR (DP14) -> FFT Accelerator (DP19) -> FFT Memory Interface (DP20) -> CPU (DP21) -> UART (DP24) -> host PC (DP26).

2.3 Combined data-path table

ID	Data path	Source	Destination	Description	Src (original IDs)
A. Bus (register / memory) data paths					
DP01	Master to AXI Interconnect	AXI Master (CPU)	AXI Interconnect	Master generates address, control and data transactions toward the interconnect.	CE DP01; SS (CPU only master)
DP02	AXI Interconnect to selected IP	AXI Interconnect	Selected IP	Address decoding routes the transaction to the selected peripheral (each in its own 4 KB slot). Unmapped or unaligned accesses return SLVERR.	CE DP02; SS
DP03	Selected IP to AXI Master (return data)	Selected IP	AXI Interconnect, then AXI Master	Read data or status from the selected IP is selected and returned to the requesting master.	CE DP03, DP17
B. Timing and acquisition data paths					
DP04	Timer counter data path	Timer control / configuration	Timer counter	Load and prescaler configuration drive the timer counter.	CE DP15; SS
DP05	Timer capture data path	Timer capture event	Timer capture register	A capture event stores the relevant timer value in the capture register.	CE DP16
DP06	Sample-tick path	Timer (compare match)	ADC Interface	Each compare match (250 Hz in the ECG configuration) triggers one conversion: one tick, one conversion.	SS step 1; CE feature 3
DP07	Analog / serial input to ADC Interface	Analog source / external ADC (analog_in, adc_vref or SPI-ADC lines)	ADC Interface Controller	Analog signal is digitized (on-chip SAR in SS, external ADC in CE/SA); serial conversion data is captured and assembled into a 12-bit sample.	SS; CE feature 4; SA stages 1-2
DP08	ADC sample data path to processor	ADC Interface / ADC FIFO (32 x 12)	AXI Master (CPU)	ADC samples are transferred through the sample/FIFO data interface for processor access. FIFO prevents overwrite when full and reports overrun.	CE DP12; SA stage 2
DP09	Hardware ADC-to-FIR stream	ADC Interface	FIR Filter	ADC pushes sample_data with sample_valid to the FIR with no bus involvement. If the FIR is busy the sample is dropped and OVERRUN is set.	SS step 2
DP10	Firmware-fed FIR input path	AXI Master (CPU)	FIR Input FIFO (8 x 16), then FIR	Samples written by firmware are buffered in the FIR input FIFO before entering FIR processing.	CE feature 6
C. FIR data paths					
DP11	FIR configuration data path	AXI Master	FIR coefficient storage	FIR coefficients are transferred through the register interface into coefficient storage; BASE / 50 Hz notch / 60 Hz notch bank selection.	CE DP10; SS; CE feature 8
DP12	FIR streaming data path	Stream input (ADC stream or FIR input FIFO)	FIR Filter	Streaming samples pass through the FIR datapath (16 or 31 taps) and filtered samples are generated (filtered_data + filtered_valid).	CE DP11; SS step 3
DP13	FIR to QRS stream	FIR Filter	FIR-to-QRS FIFO (32 x 16), then QRS Detector	Dedicated valid/ready hardware stream without AXI mediation; slow stages push back.	CE feature 9; SS steps 3-5
DP14	FIR to FFT path	FIR Filter (or ADC samples if FIR bypassed)	FFT Accelerator / FFT memory	Optionally filtered time-domain samples are supplied to the FFT stage. Exact fork from the FIR output and sample handshakes depend on the RTL.	SA stages 3-5
D. QRS / cardiac data paths					

ID	Data path	Source	Destination	Description	Src (original IDs)
DP15	ECG to QRS data path	ECG input (filtered samples)	QRS Detector	ECG samples are supplied to the QRS processing block (derivative, squaring, integration, thresholding).	CE DP06; SS step 4
DP16	QRS result data path	QRS Detector	AXI Master (CPU)	QRS processing produces R-peak, RR interval, BPM and rhythm-class results, held in double-buffered RR/HR registers read by the CPU.	CE DP07; SS
DP17	Hardware alert path	QRS Detector (alert flag)	alert_led pin	Threshold crossing drives the alert pin directly with no CPU involvement.	SS
DP18	Firmware alert path	AXI Master (CPU)	GPIO output (LED / buzzer)	Firmware reads the arrhythmia result and drives LED and buzzer outputs through GPIO.	CE feature 15
E. FFT data paths					
DP19	FFT input data path	FFT Memory Interface / data memory (or FIR output)	FFT Accelerator	A block of time-domain samples prepared by the CPU or supplied from the filter is fed to the FFT.	SA stage 4
DP20	FFT result storage path	FFT Accelerator	FFT Memory Interface / data memory	Frequency-domain results (real and imaginary components) are written through the FFT memory path.	SA stage 5
DP21	FFT result read path	FFT Memory Interface	AXI Master (CPU)	CPU retrieves frequency-bin values or components through the supported bus path and may compute magnitudes.	SA stage 6
F. Security and output data paths					
DP22	AES plaintext input path	AXI Master (CPU)	AES-128 (DATA_IN0-3, key, mode, IV)	CPU packs RR/HR (or other results) into a 128-bit block and writes key, mode, IV and block over the bus.	SS step 1 of telemetry
DP23	AES ciphertext return path	AES-128 (DATA_OUT0-3)	AXI Master (CPU)	Ciphertext is read back after DONE.	SS
DP24	UART TX data path	AXI Master (CPU)	UART TX logic -> uart_tx	Data written to the UART transmit register is transferred to the TX logic; CPU writes 16 ciphertext bytes waiting for TX_READY.	CE DP04; SS
DP25	UART RX data path	uart_rx -> UART RX logic	AXI Master (CPU)	Received serial data is captured and made available through the UART RX register (host commands).	CE DP05; SS; SA
DP26	Result path to host PC	CPU via UART	Host PC	Spectrum bins/magnitudes, QRS results or encrypted telemetry reach host software for decoding, decryption and plotting.	SA stage 7; SS
DP27	SPI TX data path	AXI Master (CPU)	SPI TX logic	Data written to the SPI TX register is transferred to the SPI transmit path (SPI2 result output).	CE DP08
DP28	SPI RX data path	SPI RX logic	AXI Master (CPU)	Received SPI data is converted to parallel data and made available through the RX register.	CE DP09
G. GPIO data paths					
DP29	GPIO input data path	GPIO input (gpio pins)	GPIO register interface	External GPIO signals enter the input logic and become readable through DATA_I.	CE DP13; SS
DP30	GPIO output data path	AXI Master	GPIO output logic -> pins	Software-written output data is transferred through GPIO output logic to the external output.	CE DP14; SS

3. Combined Control Path

The control path configures IPs, starts and paces them, reports status, and coordinates the workflow. Items 1-28 of the CARDIOEDGE control paths (CP01-CP28) are all retained, reordered into groups, and extended with the SentinelSoC and Spectrum Analyzer control behavior.

ID	Control path	Source	Destination	Description	Src (original IDs)
A. Reset, bus and handshake control					
CP01	Reset control	rst_n (and wdt_reset_n)	Interconnect and all IPs	Reset initializes the interconnect and IP control/status elements. wdt_reset_n resets the chip when the watchdog expires (scope still to be confirmed).	CE CP01; SS
CP02	AXI address control	AXI Master	AXI Interconnect	Master address is evaluated to determine the target slave.	CE CP02
CP03	AXI write control	AXI Master	AXI Interconnect / Selected IP	AW and W handshakes control acceptance of write address and write data.	CE CP03
CP04	AXI write response control	Selected IP / Interconnect	AXI Master	Write response is generated after completion of the write transaction.	CE CP04
CP05	AXI read control	AXI Master	AXI Interconnect / Selected IP	AR handshake initiates a read transaction and the selected IP supplies the response.	CE CP05
CP06	Slave selection	AXI Address	AXI Interconnect	Address decoding generates the selection for the intended peripheral (one 4 KB slot per IP). Anything outside the mapped slots returns SLVERR.	CE CP06; SS
CP07	Register access control	AXI Address and Control	Selected IP	Register offset and read/write direction determine the register operation. Unaligned access returns SLVERR; writes to read-only registers are flagged.	CE CP07; SS
CP08	AXI handshake control	AXI Master and Slave	AXI Interconnect	VALID and READY handshakes determine when AXI transfers occur.	CE CP27
CP09	Transaction routing control	AXI Masters	AXI Interconnect	Interconnect arbitration and routing determine which transaction is serviced. With the CPU as the only master there is no arbitration in the SS base.	CE CP28; SS
B. Peripheral configuration control					
CP10	UART configuration control	AXI Master	UART	Baud divisor (16-bit), line control and interrupt enable configure UART operation. TX_READY / RX_READY flags gate transmit and receive; framing and overrun errors are sticky.	CE CP08; SS
CP11	SPI configuration control	AXI Master	SPI2	SPI control, clock, slave-select and configuration registers control SPI operation.	CE CP10
CP12	SPI interrupt control	AXI Master	SPI Interrupt Logic	GIER, IER and ISR registers control and report SPI interrupt conditions.	CE CP23
CP13	ADC conversion control	AXI Master	ADC Interface	ADC control register enables and starts acquisition: START then EOC flag, programmable sample-rate divider, overrun flag when the FIR is busy. In streaming mode the Timer tick, not the CPU, paces each conversion.	CE CP12; SS
CP14	ADC interrupt control	AXI Master	ADC Interrupt Logic	IRQ enable controls ADC interrupt sources.	CE CP24
CP15	FIR configuration control	AXI Master	FIR	FIR enable, bypass mode, coefficient loading and coefficient-bank selection (BASE / 50 Hz / 60 Hz) configure the filter.	CE CP11; SS; CE feature 8
CP16	QRS processing control	AXI Master	QRS Detector	Control register configures QRS processing (adaptive threshold, refractory, low and high bpm thresholds, arrhythmia deviation threshold); status provides processing state. A configuration with LOW >= HIGH is rejected.	CE CP09; SS
CP17	Timer control	AXI Master	Timer	Control, load/reload and prescaler registers configure timer operation (periodic or one-shot, 32-bit counter, live COUNT).	CE CP19; SS
CP18	Timer compare control	AXI Master	Timer Compare Logic	Compare register determines the programmed comparison value; a match starts the next ADC conversion.	CE CP20; SS
CP19	Timer capture control	Timer Capture Logic	Timer Capture Register	Capture control determines when the timer value is stored.	CE CP21

ID	Control path	Source	Destination	Description	Src (original IDs)
CP20	Timer interrupt control	AXI Master	Timer Interrupt Logic	Interrupt enable and status registers control timer interrupt generation and clearing (cleared via IRQ_STATUS).	CE CP22; SS
C. GPIO control					
CP21	GPIO direction control	AXI Master	GPIO	GPIO direction register determines input/output behavior (per pin).	CE CP13; SS
CP22	GPIO interrupt control	AXI Master	GPIO Interrupt Logic	Interrupt enable, type, polarity and any-edge settings control GPIO interrupts; per-pin mask and pending flags (cleared via GPIO STATUS).	CE CP14; SS
CP23	GPIO output control	AXI Master	GPIO Output Logic	Set, clear, toggle and write-mask controls modify GPIO outputs.	CE CP15
CP24	GPIO debounce control	AXI Master	GPIO Debounce Logic	Debounce enable and threshold configure input filtering.	CE CP16
CP25	GPIO electrical control	AXI Master	GPIO Electrical Logic	Inversion, open-drain and pull-up/pull-down controls configure GPIO behavior.	CE CP17
CP26	GPIO PWM control	AXI Master	GPIO PWM Logic	PWM enable, configuration and duty registers control PWM operation.	CE CP18
D. Status, FIFO and streaming control					
CP27	Status feedback path	IP Status Logic	AXI Master	IP status registers provide operational state through AXI reads.	CE CP25
CP28	FIFO control and status	AXI Master / FIFO Logic	IP FIFO	FIFO control and status indicate data availability and occupancy (ADC FIFO 32x12, FIR input FIFO 8x16, FIR-to-QRS FIFO 32x16); ADC FIFO prevents overwrite when full and reports overrun.	CE CP26
CP29	Streaming valid/ready flow control	ADC, FIR, QRS stages	Next stage in the stream	Each hardware handoff is a valid/ready handshake so a slow stage pushes back instead of losing data silently. ADC drops a sample and sets OVERRUN if the FIR is busy; FIR raises a stall flag on a stuck pipeline.	SS; CE feature 9
CP30	Sticky error flags and configuration checks	ADC, FIR, QRS, UART	AXI Master (status read)	Sticky error flags on ADC, FIR, QRS and UART report faults to firmware; QRS rejects LOW >= HIGH.	SS
E. Interrupt control					
CP31	Interrupt collect / mask / merge	Interrupt sources	Interrupt Aggregator -> core	Collect: each source sets its bit in CAUSE. Filter: MASK blocks or enables each source (resets to 0x3F). Merge: enabled sources become one line to the core. Source bits: 0 Timer compare, 1 UART RX-ready, 2 QRS peak, 3 AES done, 4 Watchdog pre-timeout, 5 GPIO pin change. Core interrupt interface (single line vs vector) is still open.	SS
CP32	Interrupt priority and service	Interrupt Aggregator / ISR	CPU	ISR reads CAUSE and serves Watchdog, QRS, Timer, AES, UART, GPIO in that order.	SS
CP33	Interrupt clearing	CPU	Interrupt Aggregator and source IPs	Write 1 to each handled CAUSE bit; other bits stay. Source clears: Timer IRQ_STATUS; UART read RXDATA; QRS read RR_INTERVAL; AES STATUS; Watchdog FEED write; GPIO STATUS.	SS
CP34	Additional interrupt sources to merge	SPI2, ADC, FFT done, ext_irq[n:0]	Interrupt Aggregator	The SS aggregator covers 6 sources; SPI interrupts (CE), ADC interrupts (CE), FFT completion (SA) and external ext_irq pins must be added or routed separately. Open integration item.	CE CP23, CP24; SA; SS pins
F. Watchdog, alert and security control					
CP35	Watchdog control	AXI Master (FEED), Timer ISR	Watchdog	Programmable timeout; FEED write restarts the countdown; on a hang: pre-timeout IRQ, then wdt_reset_n. Operates even if the CPU is frozen.	SS
CP36	Hardware alert control	QRS threshold logic	alert_led	Threshold crossing sets the alert flag and alert_led follows with no interrupt or firmware. Independent of CPU state.	SS
CP37	Firmware alert control	CPU firmware	GPIO (LED / buzzer)	Firmware responds to arrhythmia events by driving LED and buzzer outputs.	CE feature 15

ID	Control path	Source	Destination	Description	Src (original IDs)
CP38	QRS peak event control	QRS Detector	Interrupt Aggregator, CPU	peak_irq on each new peak; RR/HR double-buffered so reads are never half-updated; cleared by reading RR_INTERVAL.	SS
CP39	AES operation control	AXI Master	AES-128	Load key, mode (ECB or CTR) and IV, then block into DATA_IN0-3; set START; AES sets BUSY then DONE and raises aes_done_irq. START, MODE and IV writes are ignored while BUSY.	SS
CP40	Telemetry sequencing control	CPU firmware	AES and UART	Load AES, wait for DONE, read DATA_OUT0-3, write 16 bytes to UART waiting for TX_READY.	SS
G. FFT / spectrum control					
CP41	FFT start and status control	AXI Master	FFT Control	Start bit begins processing; status register reports busy and done. CPU must not treat start as proof that results are ready. Register details to be verified against the final RTL.	SA step 4-5
CP42	FFT memory access control	CPU / FFT Accelerator	FFT Memory Interface	Governs access to FFT input and output data through a path separate from FFT Control. May use a different bus segment or protocol.	SA
CP43	Spectrum workflow sequencing	CPU firmware	ADC IF, FIR, FFT, UART	Initialize peripherals and FIR coefficients; configure ADC Interface and acquire samples (polling or interrupt); configure FIR if used; prepare FFT input and start; monitor busy/done; access results; send over UART.	SA steps 1-8
H. System sequencing					
CP44	Boot and initialization sequence	CPU firmware	All IPs	Set up UART, GPIO, Timer, ADC, FIR, QRS, AES and Watchdog, unmask interrupts, then sleep until an interrupt arrives. Each ISR is short; watchdog is fed from the timer ISR.	SS; SA step 1
CP45	Sample-rate / clock control	Timer, ADC divider, sar_clk	ADC Interface	50 MHz system clock to 250 Hz tick via Timer; ADC sample-rate divider; sar_clk (pin or internal) is still open.	CE feature 3; SS

Appendix A. Integration notes and open decisions

These are conflicts or gaps found while merging. Nothing was dropped; both versions remain in the tables above until a decision is made.

Topic	What the sources say	Needed decision / note
CPU core	CE: VeeR EL2; SS: RV32IM; SA: RISC-V CPU.	Confirm one core. VeeR EL2 is an RV32 core with the M extension, so it can satisfy the RV32IM requirement (confirm against the RTL).
Bus protocol	CE: AXI4 (mentions several masters); SS: AXI4-Lite, single master.	Pick one interconnect. Register IPs only need AXI4-Lite behavior; FFT memory traffic may need a separate or wider path.
Address map	SS defines 9 slots; slots 0x4000_8000, 0x4000_A000, 0x4001_0000 are free.	SPI2, FFT Control and FFT Memory Interface placed there as proposals only.
FIR specification	CE: 31 taps, Q1.15, 40-bit accumulator, 16-bit saturated output, 3 banks. SS: 16 taps, 12-bit in, 16-bit out, bypass.	Choose one tap count and width, or support both by configuration.
ADC front end	CE: SPI ADC + 32x12 FIFO. SS: on-chip 12-bit SAR on analog_in. SA: external ADC + interface.	Decide whether conversion is on-chip, and whether ADC pins are SPI or analog.
RR interval units	CE: BPM = 15000 / RR_INTERVAL, which fits RR counted in 250 Hz sample periods (4 ms). SS: RR in ms.	Fix one unit before writing the QRS testbench.
Sampling rate	CE: 250 Hz ECG. SA: rate depends on the analyzed signal (audio, vibration, RF).	Timer prescaler/compare must be configurable per use case.
FIR feeding mode	CE: firmware writes the FIR input FIFO. SS: ADC streams to FIR directly.	Support both with a mode select, or choose one.
FIR output fork	FIR output feeds QRS (CE, SS) and optionally FFT (SA).	A fork or selectable destination is needed; not defined in any source.
Interrupt sources	SS aggregator has 6 sources; CE adds SPI and ADC; SA adds FFT done; SS has ext_irq pins.	Extend aggregator width and CAUSE/MASK map. Core interrupt interface (single line vs vector) is open.
Open SS questions	sar_clk pin vs internal; scope of wdt_reset_n; keep AES CTR mode.	Carry forward unchanged until faculty decides.
Source numbering gap	CE feature table skips number 14.	Treated as a numbering gap, not a missing feature.
Implementation status	SA states features must be marked implemented only when RTL and tests show it; ADC may be external; spectrum display is host-side.	Add a status column (planned / RTL / verified) once the merged RTL exists.

Appendix B. Verification targets carried over

Test	Target	Src
Per-IP tests	Register read-back, UART loopback, FIR vs a Python model, AES vs NIST vectors.	SS
QRS accuracy	Detected peaks vs a labeled ECG dataset.	SS
Alert independence	Halt the CPU, inject a threshold crossing, measure latency in cycles.	SS
End to end (telemetry)	Synthetic ECG in, ciphertext out, decrypted and matched.	SS
Watchdog recovery	Hang the firmware and confirm the SoC resets cleanly.	SS
Spectrum end to end	Known multi-tone input produces expected FFT peaks at the host; verify FFT busy/done behavior against the RTL.	SA
Bus and FIFO checks	SLVERR on unmapped/unaligned access; FIFO overrun and full behavior; GPIO/Timer/SPI register behavior.	CE, SS